ADVANCED PERSISTENT THREATS

APT GROUP ANALYSIS

TARGET: ASIA

April 4, 2021
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Abstract

This article examines **Asia** as the focus of our analysis of Advanced Persistent Threat (APT) groups. We will explore the strategic objectives behind the activities attributed to these groups by examining their attack methods, targets, and operational characteristics.

Keywords: Cybersecurity, Cybercrime, APT, Threat, Asia, China, North Korea

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INTRODUCTION

When discussing APT groups attributed to Asia, the first countries that come to mind are **China** and **North Korea**. However, the Asian intelligence landscape extends far beyond these two nations, as cyber operations have also been attributed to countries such as **South Korea, Pakistan, Lebanon, and Vietnam**.

APT groups operating in Asia frequently target industries of strategic importance, including:

- Financial services
- Defense
- Energy
- Research and Development (R&D)
- Healthcare
- Telecommunications
- Government agencies

Their motivations generally fall into three categories:

- Financial gain
- Strategic intelligence collection
- Supporting the geopolitical interests and international influence of the sponsoring state

Analyzing every country individually would require an extensive study. Therefore, this article focuses on the two most significant cyber powers in the region: **North Korea** and **China**.

NORTH KOREA

North Korea is particularly difficult to analyze due to the secrecy surrounding its political system. Nevertheless, valuable information has emerged through defectors, official statements issued by the regime itself, and independent attribution performed by cybersecurity researchers.

On **February 1**, Microsoft and Google reported an espionage campaign targeting vulnerability researchers. The operation was attributed to the North Korean APT group **Zinc**, more commonly known as **Lazarus**, which deployed custom-built malware specifically developed for this campaign.

The development of tailor-made malware has become one of Lazarus' defining characteristics. The group has been active since at least **2009**, with its operational tempo increasing significantly after **2015**.

Among the operations commonly attributed to Lazarus are:

- The **WannaCry** ransomware outbreak (2017)
- Cyberattacks against **Korea Hydro & Nuclear Power (KHNP)**
- The **Sony Pictures Entertainment** breach (2014)

The attribution of the Sony Pictures attack was made by the **FBI**. North Korea subsequently denied responsibility and even offered to cooperate with the United States in investigating the incident.

Initially, the **2016 SWIFT banking attacks** were also attributed to Lazarus. Given North Korea's previous involvement in financially motivated operations—including attacks against the **Bangladesh Bank** and **Cosmos Bank of India**—this attribution initially appeared plausible.

However, following the publication of classified NSA cyber tools by the **Shadow Brokers**, researchers suggested that some of those attacks were instead connected to **Equation Group**, an organization widely believed to be associated with the U.S. National Security Agency (NSA).

More recently, South Korean intelligence agencies have claimed that North Korea has increasingly relied on the theft of cryptocurrencies to finance its nuclear weapons program.

CHINA

Attempting to examine every APT group attributed to China would require an entire book. Consequently, this article focuses only on several of the most representative groups.

According to multiple intelligence reports, **APT1** and **APT2** are associated with Units **61398** and **61486** of the People's Liberation Army (PLA).

Unlike many other APT attributions, China's involvement goes beyond mere speculation. In **2013**, Chinese authorities publicly acknowledged the existence of secret cyberwarfare units composed of both military personnel and civilians, although no operational details were disclosed.

These groups are believed to target organizations in:

- The United States
- Europe
- Japan

particularly those operating in the aerospace and defense sectors.

Another notable actor is **APT31 (Zirconium)**, whose primary mission appears to be cyber espionage and intelligence collection against foreign governments and intelligence agencies.

Researchers believe this assessment is supported by the group's use of an NSA **Equation Group** zero-day exploit several years before that exploit became publicly available through the Shadow Brokers leak in 2017.

APT10 AND APT41

Among the few Chinese APT groups for which information about individual members has become publicly available are **APT10** and **APT41**.

Several members of these groups have been identified and placed on the **FBI's Cyber Most Wanted** list. Although their activities are believed to be linked to Chinese interests, publicly available evidence has primarily resulted in criminal indictments against specific individuals rather than definitive proof of direct state control.

Both groups have been associated with long-running cyber espionage campaigns targeting governments, multinational corporations, technology companies, healthcare organizations, and telecommunications providers around the world.

WHAT IS THE RELATIONSHIP BETWEEN NORTH KOREA AND CHINA?

From a geopolitical perspective, China and North Korea have traditionally maintained close diplomatic relations. Besides sharing a border, both countries have historically operated under similar political systems.

For China, maintaining North Korea as a strategic ally offers several advantages. If North Korea were to develop a close military relationship with the United States—as briefly seemed possible during the meetings between Donald Trump and Kim Jong-un—American military forces could potentially be stationed directly along China's northeastern border.

This strategic concern explains some of China's policies toward North Korea, including the repatriation of North Korean defectors who successfully cross into Chinese territory.

Conversely, a stronger military and intelligence partnership between China and North Korea could create a united regional front capable of challenging U.S. influence throughout East Asia.

Following the election of President Joe Biden, however, relations between the United States and North Korea cooled considerably compared with the diplomatic engagement seen during the previous U.S. administration.

ARE THERE KNOWN COLLABORATIONS BETWEEN THEIR APT GROUPS?

Although joint cyber operations between China and North Korea cannot be completely ruled out, there is currently little public evidence supporting sustained operational cooperation between their respective APT groups.

The nature of Advanced Persistent Threat operations generally makes such collaboration unlikely.

APT groups are primarily tasked with:

- Intelligence collection
- Strategic espionage
- Supporting national security objectives
- Generating financial resources for their sponsoring organizations

These missions frequently require operating independently—even against countries considered political allies—because each nation prioritizes its own intelligence requirements.

As a result, even governments with friendly diplomatic relations may conduct cyber espionage against one another.

ARE THERE OTHER ASIAN APT GROUPS?

Absolutely.

Several other Asian countries have been linked to Advanced Persistent Threat groups, including:

- **DarkHotel**, attributed to South Korea.
- **OceanLotus** (also known as **APT32**), commonly attributed to Vietnam.

However, compared with the volume of publicly attributed operations linked to China and North Korea, these groups represent a much smaller portion of the known Asian APT landscape.

WHAT ABOUT JAPAN?

Despite being one of the world's leading technological powers, **no APT group has yet been publicly attributed to Japan**.

This does **not** necessarily mean that Japanese cyber operations do not exist.

It is entirely possible that:

- Japanese cyber capabilities have not been publicly identified.
- Their operations remain classified.
- Their priorities differ from those of neighboring countries, focusing less on traditional cyber espionage and more on areas such as organized crime investigations, domestic intelligence, or defensive cybersecurity.

The absence of public attribution should therefore not be interpreted as evidence that Japan lacks advanced cyber capabilities.

CONCLUSION

Asia remains one of the world's most active regions in terms of Advanced Persistent Threat activity.

China and North Korea dominate the landscape due to the number, sophistication, and persistence of the groups attributed to them. Their operations consistently target sectors of strategic importance—including defense, finance, research, critical infrastructure, and government institutions—with objectives ranging from intelligence gathering to financial gain and geopolitical influence.

Although public attribution provides valuable insights into these operations, definitive identification of the organizations responsible remains exceptionally difficult. Attribution is ultimately based on technical evidence, behavioral analysis, and intelligence assessments rather than absolute proof.

Understanding these groups, their methods, and their strategic objectives is essential for improving global cybersecurity and strengthening defenses against increasingly sophisticated cyber threats.

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